

Project Design Phase

Problem – Solution Fit Template

Date	01 October 2026
Team ID	SWTID-2026-1665
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Template:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS - School and college students (all academic levels) who study on their own laptops/PCs, e.g. Mac M1 - Self-learners picking up new skills such as SQL - Learners with limited or no internet access at home, hostel or college	6. CUSTOMER CONSTRAINTS CC - Poor, unstable or costly internet / mobile data - Limited budget: cannot afford paid tutors or subscriptions - Modest device hardware and storage, so the model must be lightweight - Privacy worries about uploading study material to the cloud - Little time before exams	5. AVAILABLE SOLUTIONS AS - Textbooks, notes, PDFs: work offline but are static; no doubt-clearing or quizzes - Tuition / coaching: personal but costly, fixed timings - Google search, YouTube: need internet, scattered information - Online AI chatbots and learning apps: need internet, subscriptions, data sharing - Differentiator: EduGenie works offline and combines Q&A, simplified explanations, quizzes, summaries and learning paths in one tool	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P - Get quick, clear answers to doubts - Understand complex concepts in simple language - Check their level of understanding (e.g. Pythagoras Theorem) through quizzes - Summarize long educational passages - Follow a structured beginner-to-advanced plan for a new topic (e.g. SQL)	9. PROBLEM ROOT CAUSE RC - Study material is static and generic, not matched to the learner's level - No instant help for doubts outside class hours - Most AI study tools depend on a constant internet connection - No easy way to self-assess understanding - No guided roadmap, so topics are studied randomly - Long chapters are hard to digest quickly	7. BEHAVIOUR BE - Ask teachers or friends, or search online when connectivity allows - Rely on textbooks, handwritten notes, downloaded PDFs and previous question papers when offline - Study in bursts before exams - Try practice questions or mock tests to check progress - Use free tools and prefer simple, quick-to-use apps	
Identify strong TR & EM	3. TRIGGERS TR - An upcoming exam, test or assignment - Not understanding a topic taught in class - A lengthy chapter that must be revised quickly - Wanting to learn a new subject or skill (e.g. SQL) - Internet outage while studying	10. YOUR SOLUTION SL - EduGenie: a lightweight, offline-based AI learning assistant that runs locally on the learner's own device (FastAPI backend + simple HTML & CSS frontend), works well on devices like Mac M1 - Ask questions and get smart, concise answers - Simplified explanations of complex concepts - Quiz generation from a topic or text to test understanding - Personalized learning paths with beginner-to-advanced topics and timelines - Summaries of large passages - No internet needed for everyday use, no subscription, and study data stays on the device - Can be extended with multilingual learning, voice assistance, progress tracking and adaptive quizzes	8. CHANNELS: 8.1 ONLINE CH - Limited and occasional use only - One-time download and setup of EduGenie and its model - Searching topics on Google / YouTube and downloading study PDFs when connected	Extract online & offline CH or BE
	4. EMOTIONS: BEFORE / AFTER EM - Before: confused, anxious, overwhelmed, dependent on internet, unsure of progress - After: confident, in control, independent, motivated and relieved		8.2 OFFLINE CH - Main channel: EduGenie runs locally on the learner's device anytime, anywhere, without internet - Classrooms, college labs and study groups - Sharing the app with classmates through pen drive or local network - Teachers, mentors, workshops and webinars for demos and feedback	